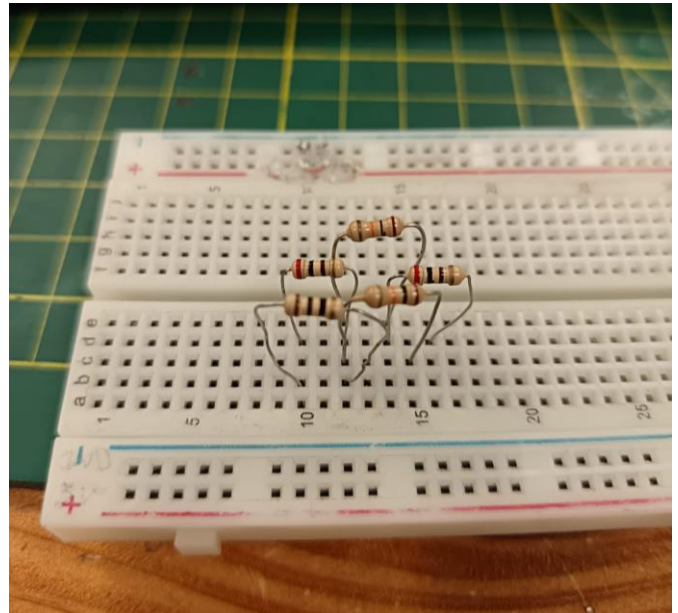
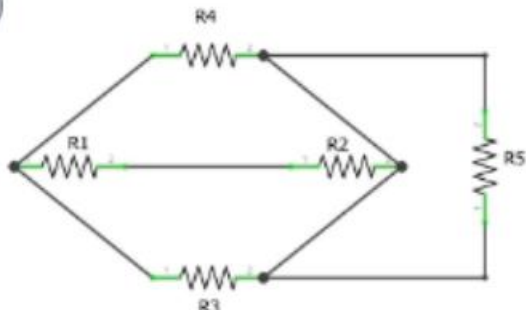


RETO N° 1. ARREGLO DE RESISTENCIAS

Implemente los siguientes circuitos en el protoboard y calcule la **resistencia equivalente**.

1



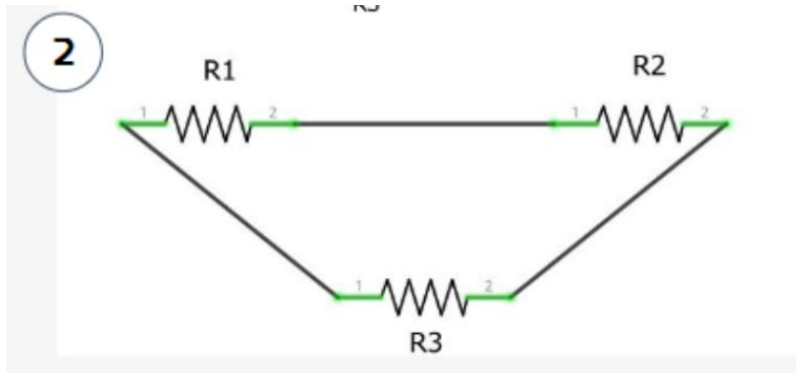
RESOLUCIÓN:

① $R_1 = 220 \text{ ohmios}$
 $R_2 = 4700 \text{ ohmios}$
 $R_3 = 10000 \text{ ohmios}$
 $R_4 = 1000 \text{ ohmios}$
 $R_5 = 220 \text{ ohmios}$

Resistencia en paralelo ; $R_1 + R_2 = R_{12} = 4920$

$$R_{eq} = \frac{1}{\frac{1}{R_{12}} + \frac{1}{R_3} + \frac{1}{R_4} + \frac{1}{R_5}}$$

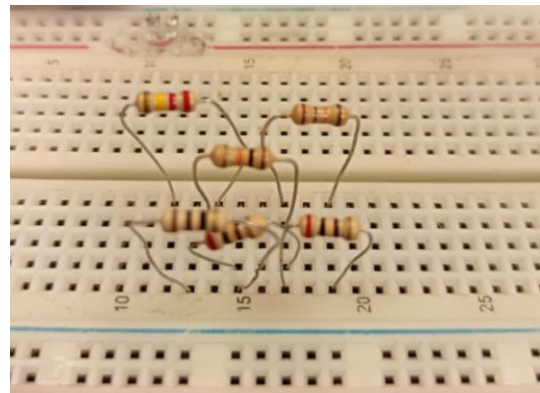
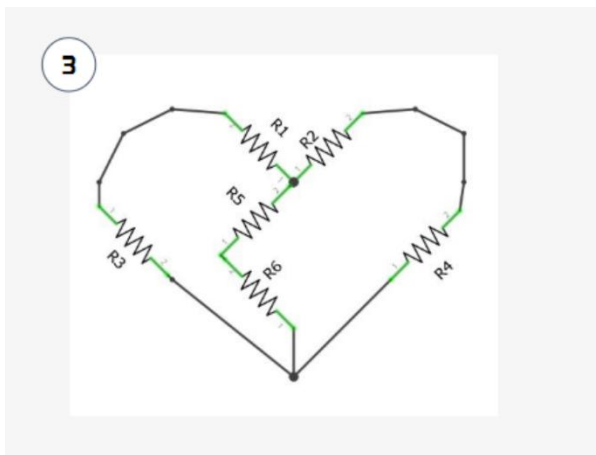
$$R_{eq} = \frac{1}{\frac{1}{4920} + \frac{1}{10000} + \frac{1}{1000} + \frac{1}{220}} = 170,98 \text{ ohmios (teórico)}$$



RESOLUCIÓN:

2

$R_1 = 220 \text{ ohmios}$ $R_2 = 4700 \text{ ohmios}$ $R_3 = 10000 \text{ ohmios}$	$\left. \begin{array}{l} R_1 \\ R_2 \\ R_3 \end{array} \right\} \text{Resistencias en serie}$
	$R_{eq} = R_1 + R_2 + R_3$
	$R_{eq} = 220 \text{ ohmios} + 4700 \text{ ohmios} + 10000 \text{ ohmios}$
	$R_{eq} = 14920 = \underline{14.92 \text{ k ohmios}} \text{ (teórico)}$



RESOLUCIÓN:

3

$R_1 = 220$	$\left. \begin{array}{l} R_1 = 220 \\ R_2 = 220 \\ R_3 = 1000 \\ R_4 = 1000 \\ R_5 = 330 \\ R_6 = 330 \end{array} \right\}$	Resistencia en Paralelo:	$R_1 + R_3 = R_{13} = 1220$
$R_2 = 220$		$R_2 + R_4 = R_{24} = 1220$	
$R_3 = 1000$		$R_5 + R_6 = R_{56} = 660$	
$R_4 = 1000$			
$R_5 = 330$			
$R_6 = 330$			

$$\frac{1}{R_{eq}} = \frac{1}{R_{13}} + \frac{1}{R_{24}} + \frac{1}{R_{56}}$$

$$R_{eq} = \frac{1}{\frac{1}{R_{13}} + \frac{1}{R_{24}} + \frac{1}{R_{56}}} = \frac{1}{\frac{1}{1220} + \frac{1}{1220} + \frac{1}{660}} = 317,007$$

~~Req = 317,007 ohms~~

$R_{eq} = 317,007 \text{ ohms}$